

Proof. **Step1. Degenerate / edge cases.**

1. **Case $n = 1$.** Since $\sum_{i=1}^1 \lambda_i x_i = x_1$ and $\sum_{i=1}^1 \lambda_i f(x_i) = f(x_1)$, inequality (J) reads $f(x_1) \leq f(x_1)$, which holds with equality.
2. **Zero weights.** If $\lambda_k = 0$ for some index k , the term $\lambda_k f(x_k)$ vanishes and the left-hand side does not contain x_k . Hence the inequality for the full family $\{x_i, \lambda_i\}$ is equivalent to the inequality for the reduced family obtained by deleting all indices with zero weight. Therefore it suffices to prove (J) under the assumption

$$\lambda_i > 0 \quad \text{for all } i.$$

3. **A weight equal to 1.** If $\lambda_j = 1$ (and consequently all other $\lambda_i = 0$), then $\sum_i \lambda_i x_i = x_j$ and $\sum_i \lambda_i f(x_i) = f(x_j)$; again (J) becomes an equality.

Thus, apart from these trivial situations, we may assume

$$n \geq 2, \quad \lambda_i > 0 \quad (i = 1, \dots, n). \quad (1)$$

Step2. Base case $n = 2$.

For two points x_1, x_2 and a weight $\lambda \in [0, 1]$ (so the second weight is $1 - \lambda$), the definition of convexity gives

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2). \quad (2.1)$$

Equality in (2.1) holds *iff* $\lambda \in \{0, 1\}$ or f is affine on the segment $[x_1, x_2]$ (in particular when $x_1 = x_2$). Hence (J) is true for $n = 2$.

Step3. Inductive hypothesis.

Assume that for some integer m with $2 \leq m \leq n - 1$ the inequality (J) holds for any collection of m points and any non-negative weights summing to 1; i.e.

$$f\left(\sum_{i=1}^m \mu_i x_i\right) \leq \sum_{i=1}^m \mu_i f(x_i) \quad (\mu_i \geq 0, \sum_{i=1}^m \mu_i = 1). \quad (3.1)$$

Step4. Inductive step $m = n - 1 \implies m = n$.

Let the weights $\lambda_1, \dots, \lambda_n$ satisfy (1). The case $\lambda_1 = 1$ is covered by Step1.3, so we may assume

$$0 < \lambda_1 < 1.$$

Define

$$y := \frac{\sum_{i=2}^n \lambda_i x_i}{1 - \lambda_1}.$$

Because

$$\sum_{i=2}^n \frac{\lambda_i}{1 - \lambda_1} = 1,$$

the numbers

$$\mu_i := \frac{\lambda_i}{1 - \lambda_1}, \quad i = 2, \dots, n,$$

form a legitimate set of weights for the points $x_2, \dots, x_n$.

4.1 Apply the inductive hypothesis to $\{x_2, \dots, x_n\}$. Using (3.1) with the points $x_2, \dots, x_n$ and the weights μ_i we obtain

$$f(y) = f\left(\sum_{i=2}^n \mu_i x_i\right) \leq \sum_{i=2}^n \mu_i f(x_i) = \frac{1}{1 - \lambda_1} \sum_{i=2}^n \lambda_i f(x_i). \quad (4.1)$$

Multiplying (4.1) by the positive factor $1 - \lambda_1$ yields

$$(1 - \lambda_1) f(y) \leq \sum_{i=2}^n \lambda_i f(x_i). \quad (4.2)$$

4.2 Convexity step for the pair (x_1, y) . Applying the two-point convexity (C) with weight λ_1 gives

$$f(\lambda_1 x_1 + (1 - \lambda_1)y) \leq \lambda_1 f(x_1) + (1 - \lambda_1)f(y). \quad (4.3)$$

4.3 Identify the aggregated point. By the definition of y ,

$$\lambda_1 x_1 + (1 - \lambda_1)y = \lambda_1 x_1 + \sum_{i=2}^n \lambda_i x_i = \sum_{i=1}^n \lambda_i x_i. \quad (4.4)$$

Thus the left-hand side of (4.3) is exactly $f(\sum_{i=1}^n \lambda_i x_i)$.

4.4 Assemble the inequalities. Insert (4.2) into the right-hand side of (4.3) and use (4.4):

$$\begin{aligned} f\left(\sum_{i=1}^n \lambda_i x_i\right) &\leq \lambda_1 f(x_1) + (1 - \lambda_1)f(y) \\ &\leq \lambda_1 f(x_1) + \sum_{i=2}^n \lambda_i f(x_i) = \sum_{i=1}^n \lambda_i f(x_i). \end{aligned}$$

Hence (J) holds for n points, completing the induction.

Step5. Equality condition.

Equality in (J) occurs precisely when equality holds in both (4.3) and (4.2).

- *Equality in (4.3).* By the definition of convexity, equality holds iff either $\lambda_1 \in \{0, 1\}$ (the trivial weight case) or f is affine on the segment joining x_1 and y (in particular when $x_1 = y$).
- *Equality in (4.2).* Equality in the inductive hypothesis (4.1) holds iff either one of the weights $\lambda_2, \dots, \lambda_n$ equals 1 (the rest being 0), or f is affine on the convex hull of $\{x_2, \dots, x_n\}$ (in particular when all those points coincide).

Combining these observations and recalling that the cases where a weight equals 1 have already been dealt with in Step1, we obtain the following complete description:

1. If at least one weight equals 1 (and the others are zero), equality holds (trivial case).
2. Otherwise all weights are strictly positive. Equality holds *iff* the points with positive weight are all identical, i.e.

$$x_1 = x_2 = \dots = x_n,$$

or more generally iff f is affine on the convex hull of the points $\{x_i\}$. For a strictly convex function the only non-trivial equality case is the former.

Step6. Conclusion. We have proved Jensen's inequality (J) for any finite collection of real numbers and any convex function, treated all degenerate situations explicitly, and gave a rigorous derivation of the equality cases.

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